

CHEM 2312 LS

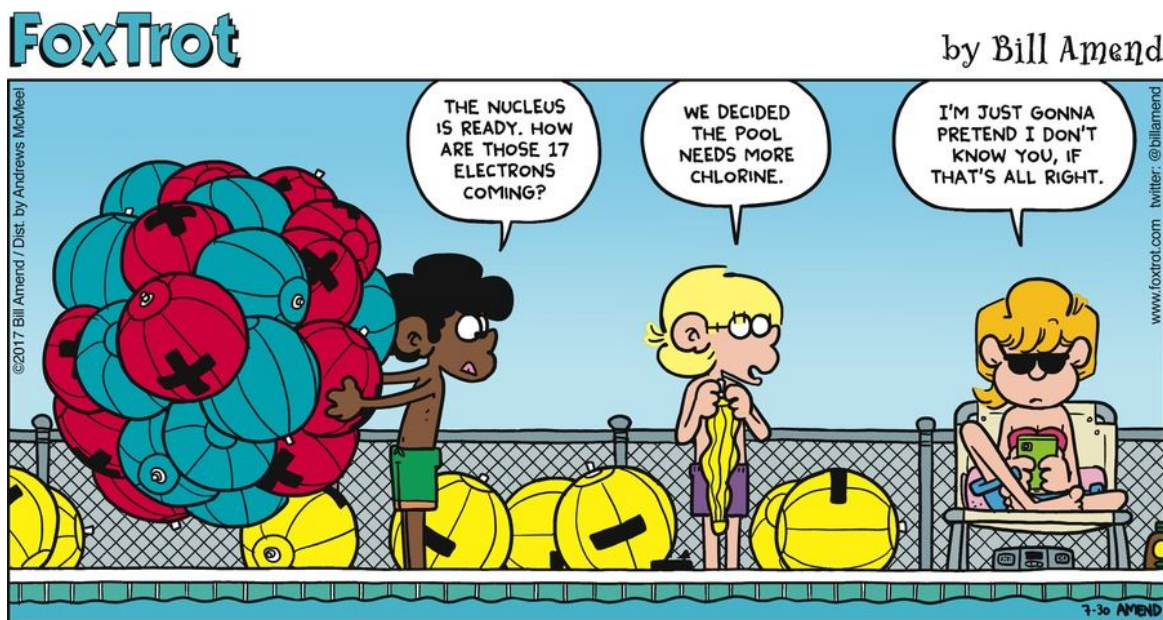
Organic Chemistry II

Midterm Examination 2

Take a deep breath. You got this.

Do not open this exam until you are instructed to do so. Only a pencil or pen, scratch paper, and molecular model kit are permitted. You will have 50 minutes to complete the exam; it can be submitted at any time after 25 minutes have elapsed. Write all responses clearly in the boxes or spaces provided; work outside of designated areas will not be graded. On multiple-choice questions, write the letter of your choice in the box *and* circle your choice.

Ensure that configurations can be unambiguously assigned at all stereocenters (*R/S* or *E/Z*). Use “+ En” to indicate the presence of a racemic mixture when chiral products are generated from achiral starting materials.

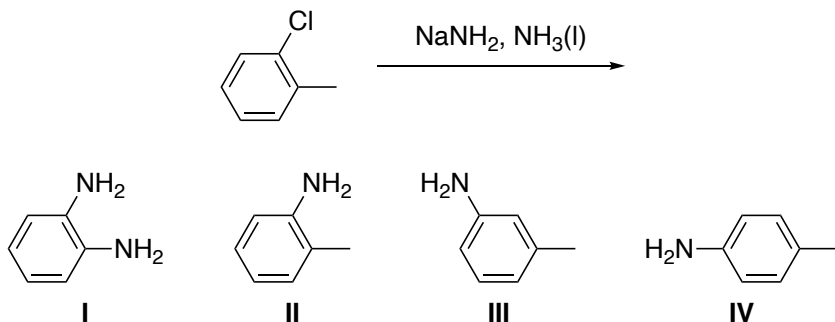


Name

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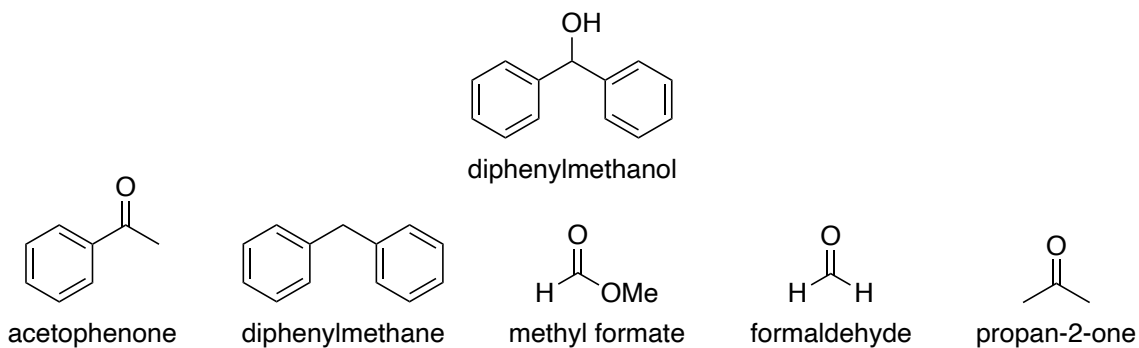
Multiple-choice Questions. Select the best response. Circle your choice and write its letter in the box. Each question is worth **4 points**.

1. Which of the following product(s) is/are *NOT* formed in the reaction below?

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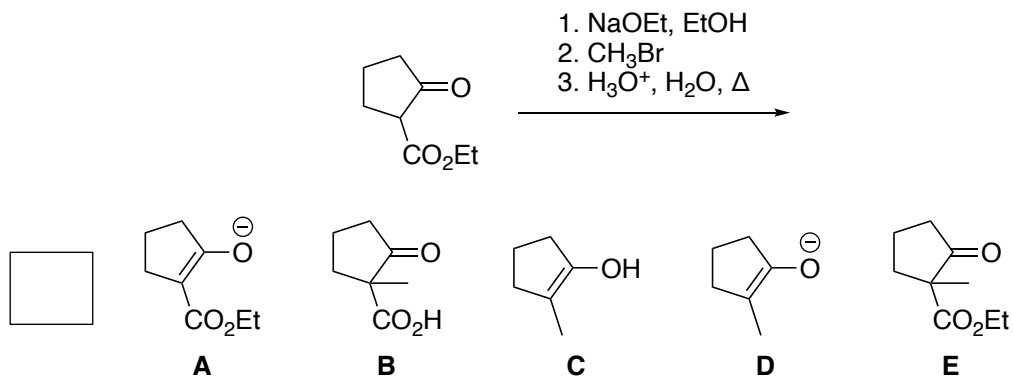
- A. **I** only
B. **II** only
C. **III** only
D. **IV** only
E. **I** and **IV**
F. **I, III, and IV**

2. Which of the following compounds could be used as a starting material to synthesize diphenylmethanol via a Grignard reaction?

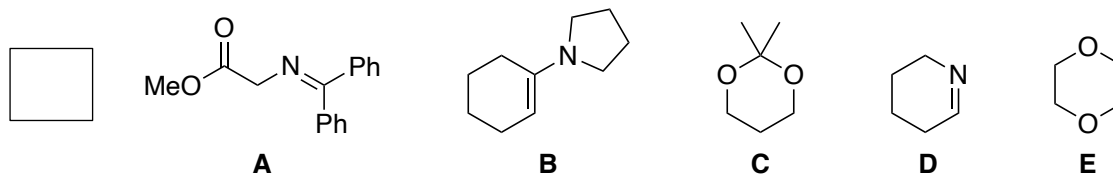
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- A. Acetophenone
B. Diphenylmethane
C. Methyl formate
D. Formaldehyde
E. Propan-2-one

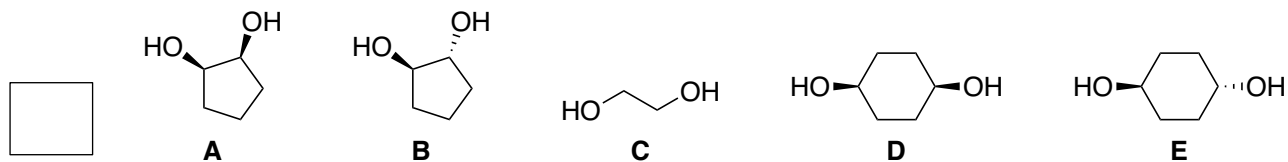
3. Which of the following structures is *NOT* involved in the sequence of reactions below as a product or reactive intermediate?



4. Of the following compounds, only one is unreactive when treated with acidic water. Select this compound.



5. Which of the following diols would give a racemic mixture of enantiomers upon reaction with one equivalent of Ac_2O and pyridine? Assume only one hydroxyl group reacts in each case.

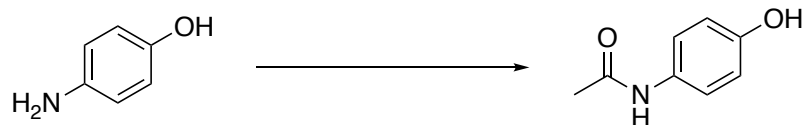


Other Questions. Place your response in the designated space for each question.

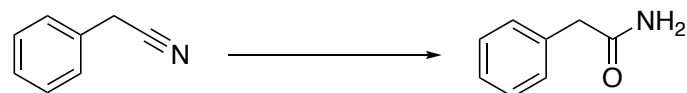
6. **12 pts.** Add reagents that would accomplish each transformation below above the arrow. Each transformation should be done in a single step.



B.

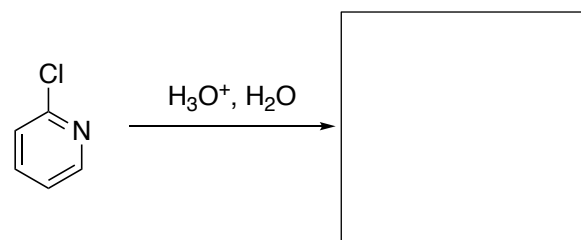


C.

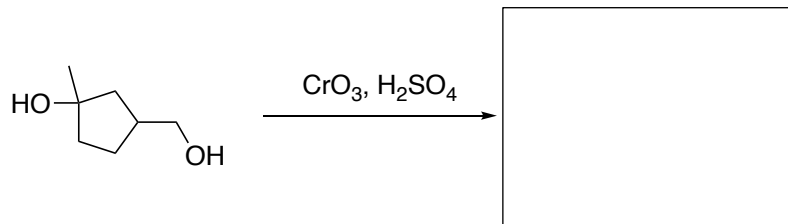


7. **36 pts.** Predict the major organic product of each reaction or sequence of reactions below. If multiple isomers are possible, draw only the major product.

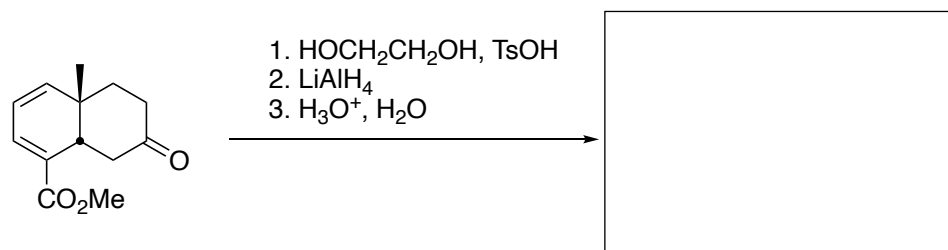
A.



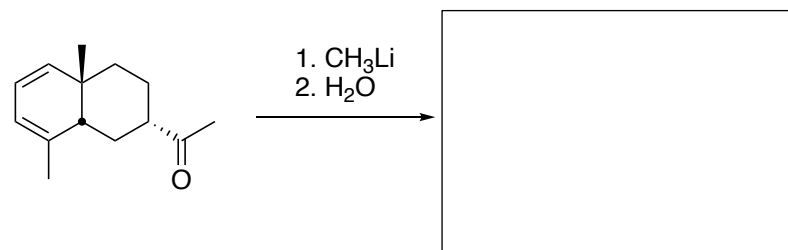
B.



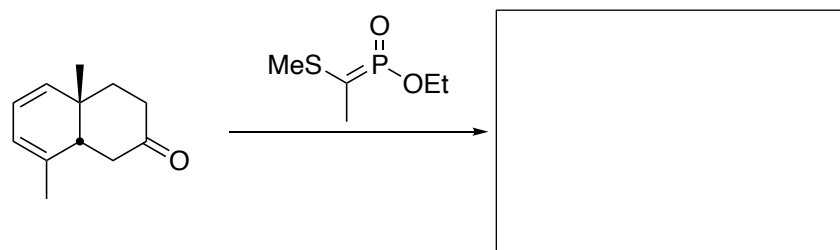
C.



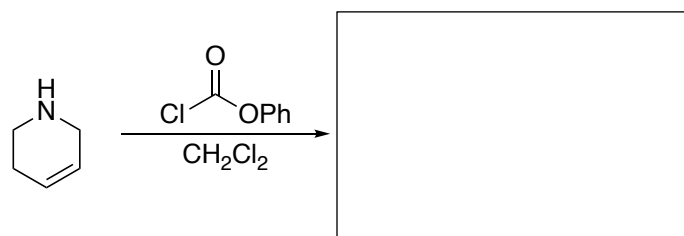
D.



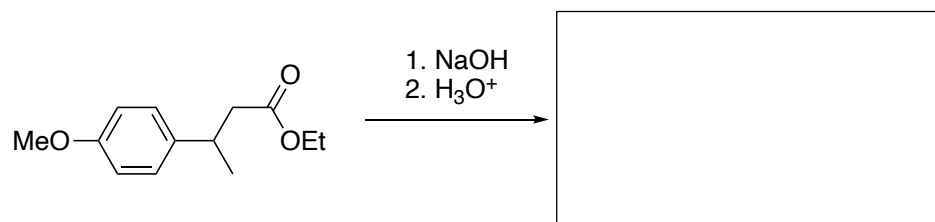
E.



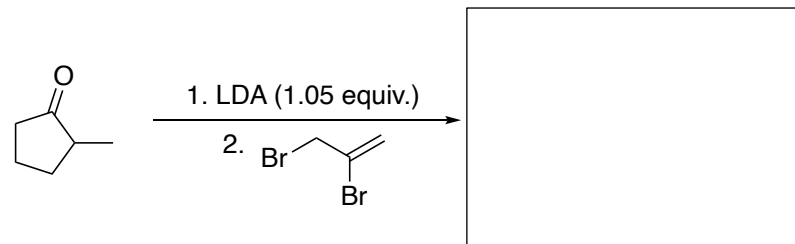
F.



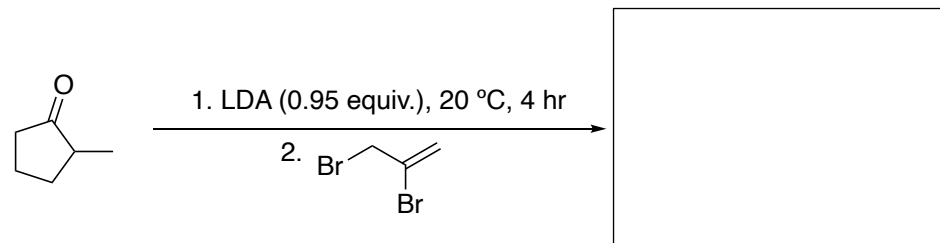
G.



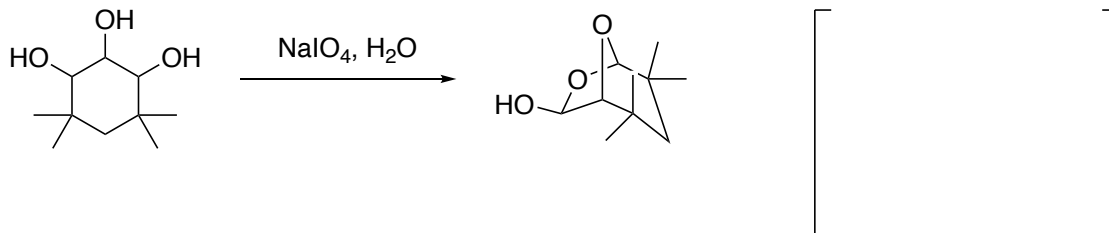
H.



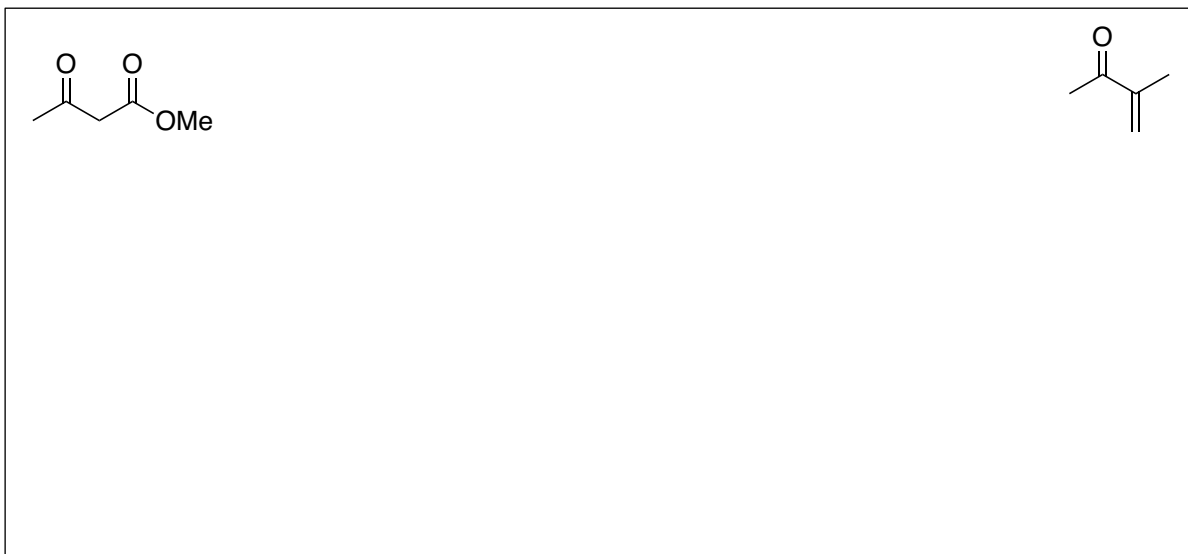
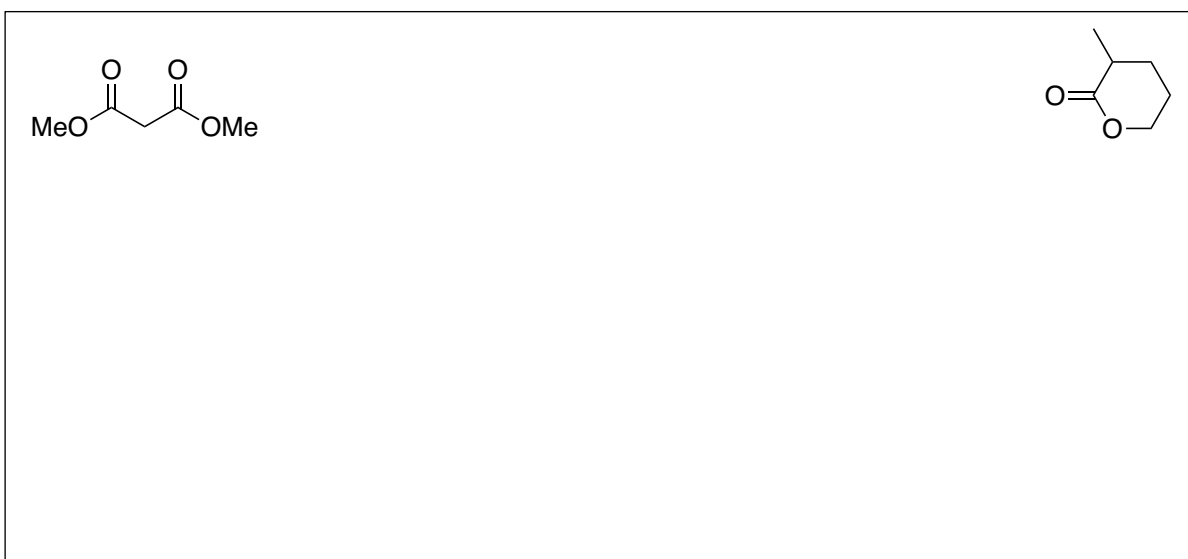
I.



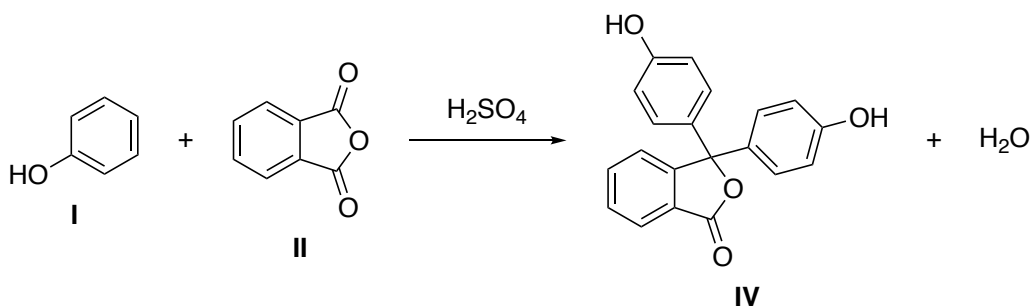
8. **8 pts.** Periodate oxidation of the triol below gave the product shown.



- A. On the product structure above, circle the hemiacetal carbon and draw a box around the acetal carbon.
- B. In the brackets above right, draw the structure of a compound with two aldehyde groups and a hydroxyl group that would cyclize to give the product.
9. **12 pts.** Design a synthesis of each of the following targets (right) from the indicated starting materials (left). You are welcome to use any reagents and sources of carbon to accomplish the synthesis, but they must start from the given starting materials.



10. **12 pts.** The mechanism of the reaction below involves two stages. In the first stage, Friedel-Crafts acylation converts one equivalent of **I** and **II** to intermediate **III** (see the box below). In the second stage, **III** condenses with a second equivalent of **I** to form the product **IV** and H_2O .



Draw a mechanism with curved arrows and all reactive intermediates for the conversion of **III** and **I** to **IV** and H_2O *omitting proton transfers* (just write “pt” over the arrow and transfer protons as needed). Once electron flow involving **I** is done, it can be abbreviated “Ar.”

